

Combined Mean

If $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_k$ are the mean of k component series of size $n_1, n_2, \dots, n_k$ respectively, then mean $\bar{x}$ of the composite series is

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + \dots + n_k\bar{x}_k}{n_1 + n_2 + \dots + n_k}$$

$$\bar{x} = \frac{\sum_{i=1}^k \bar{x}_i n_i}{\sum_{i=1}^k n_i}$$

Combined Mean

If a series of an observation consists of two or more component series the mean of the whole series can be easily expressed in terms of the mean of the component series.

$$\bar{X} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$$

where $\bar{x}_1$ and $\bar{x}_2$ are the means of the combined distribution n_1 and n_2 are the total frequencies of the component distribution.

- ① There are 60 students in I year. Boys which 25 are girls. The average weight of 25 girls is 40 kg and that of 35 boys is 53 kg. Find the mean weight in kg of the entire I year.

47.583

- ② The mean age of 40 students is 16 years, and the mean age of 60 students is 20 years. Find out the mean age of all 100 students combined together.

18.4

- ③ A cooperative bank has two branches employing 50 and 70 workers respectively. The average salaries paid by two respective branches are ₹ 360 and ₹ 390 per month. Calculate the mean of the salaries of all the employees.

377.5

4) The mean monthly ~~sat~~ salary paid to all employees in company is ₹ 1300. The male and female employees' monthly average salaries are ₹ 1340 and ₹ 1260 respectively. Determine the percentage of males and females employed by the company.

50 percent

5) Howrah Vivekanand Institution has 4 sections in class X, having 40, 35, 45 and 42 students. The mean marks obtained in a chemistry test are 50, 60, 55 and 45 respectively in 4 sections. Determine the over all average of the marks per student.

52.25

Weighted Mean

Consider $x_1, x_2, \dots, x_n$ are variables with the weight $(w_1, w_2, \dots, w_n)$ then

$$\bar{x} = \frac{x_1 w_1 + x_2 w_2 + x_3 w_3 + \dots + x_n w_n}{w_1 + w_2 + \dots + w_n}$$

$$\Rightarrow \bar{x} = \frac{\sum xw}{\sum w}$$

example:-

An investor is fond of investing in equity shares. During a ~~period~~ period of falling prices in the stock exchange, a stock is sold at Rs 120 per share on day 1, Rs 105 on the next and Rs 90 on the third day. The investor has purchased 50 shares on the first day, 80 shares on the second day and 100 shares on the third day. What average price per share did the investor pay?

Limitations of statistics

- ① Statistics does not study qualitative phenomenon
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1 Limitation of Statistics

- ① Statistics is not study qualitative phenomenon
Statistics, being a science dealing with of numerical data, is applicable to the study of only those subject of enquiry which are capable of quantitative measurement. As such qualitative phenomena like honesty, poverty, culture ~~are~~ etc. which cannot be expressed numerically, are not capable of direct statistical analysis.

- 2) Statistics does not study individuals:-
Statistics deals with an aggregate of objects and does not give any specific recognition to the individual items of a series. Individual items, taken separately, do not constitute statistical data and are meaningless for any statistical enquiry.

- 3) Statistical laws are not exact.
Unlike the laws of physical and natural sciences, statistical laws are only approximations and not exact. On the basis of statistical analysis we can talk only in terms of probability and chance and not in terms of average.

② Statistical is liable to be misused

Perhaps the most important limitation of Statistics is that it must be used by experts. The use of statistical tools by inexperienced and untrained persons might lead to very fallacious conclusions.